

Xin (Sean) Zhou

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WORK EXPERIENCE

AI Engineer

2026.05 – present

Independent Contractor

Melbourne, VIC

Ordering and operations platform for To Be Yabby, Melbourne[🔗]. Built the AI agent stack.

- Designed and delivered a **multi-agent AI system** on one shared **Pipecat** voice pipeline, Silero VAD to AssemblyAI **STT** to Claude to Fish Audio **TTS**, serving a customer agent over Twilio and an authenticated staff agent over **WebRTC**, each behind its own isolated tool registry as the security boundary.
- Built the **agentic tool-calling layer** that lets the assistant do the operator's job rather than describe it, a **ReAct loop** over **16 LLM function-calling tools** across orders, stock, menu and daily takings.
- Enforced **human approval** on every order an agent creates, reading back items, total and payment before writing it; withheld irreversible actions from every role including the owner; and hardened both agents against **prompt injection**, with caller identity taken from signed **JWT** claims rather than model input.
- Delivered **multi-modal AI order intake**, turning customer message screenshots into structured orders through **Claude vision** extraction, customer resolution, menu matching and automated form completion.
- Architected a **delegated UI-automation sub-agent** for voice-driven dashboard control, reading the page's **accessibility tree** and driving the matching controls on a smaller model chosen for latency.
- Built the platform's **role-based permission system**, role templates plus per-user grants enforced per request across **271 endpoints**, and gated every agent tool through it so an agent inherits its operator's authority.
- Rebuilt the customer phone line as a **conversational voice AI agent** over a bidirectional Twilio WebSocket, with **barge-in interruption**, tuned end-of-turn detection, and menu-derived keyterms for bilingual speech, retiring a request-response design the caller could not interrupt.
- Delivered **multi-turn session continuity** across mid-sentence network drops, rejoining the same **LiveKit room** with transcript and tool context intact, and every session persisted as one **auditable tool-call trace**.
- Engineered a custom client-side **LiveKit transport** behind Pipecat's transport interface, keeping **RTVI and the client hooks above it unchanged**, shipped behind a feature flag and the incumbent then retired.
- Filed an upstream **issue and pull request** on **Pipecat's LiveKit transport**[🔗], with a reproduction harness; the framework's own fix cites it, and adopted a second defect from the review[🔗].

Software Engineer · Intern

2024.11 – 2025.10

Museums Victoria

Melbourne, VIC

Great Melbourne Telescope restoration at Scienceworks Museum. Built the AI document retrieval system.

- Architected a **hybrid RAG system** over an engineering design archive, combining semantic chunking, BM25 keyword and dense embedding retrieval, and rank fusion, achieving **95% top-5 accuracy at 8 ms query latency**.
- Built a **grounded question-answering service** with per-sentence citation, abstention on out-of-scope queries, and **MCP integration** exposing three retrieval tools to coding agents, served through a FastAPI console.
- Engineered a **LangGraph verification agent** adjudicating documentation claims against source code as ground truth, with **conditional-edge retry**, disconfirming-evidence search, and a bounded two-round search loop.

PROJECT EXPERIENCE

StyleAI

2025.03 – 2025.07

Personal styling system built by a four-person team. Co-designed the pipeline and owned deployment.

- Co-designed a **multi-modal styling pipeline**, a **vision LLM** to **DistilBERT** attribute embeddings to a Fashn.ai **generative virtual try-on**, turning a single user photo into a rendered outfit.
- Architected the retrieval stage as **attribute-level matching** over **sentence embeddings**, scoring each attribute independently rather than one vector per image, then a **re-ranking pass** to a single result.
- Deployed the system self-hosted, **Docker** services behind **nginx** and an outbound-only **Cloudflare Tunnel**, running inference as polled background jobs so slow model calls stay off the request path.

TECHNICAL SKILLS

Languages	Python, TypeScript, JavaScript
AI & Agentic Systems	Multi-agent architecture, RAG / hybrid retrieval, MCP, LangGraph, Multi-modal vision
Real-Time Voice	WebRTC, WebSocket, Streaming STT / TTS, LiveKit
Web Development	FastAPI, PostgreSQL, Next.js, React, Server-Sent Events
Infrastructure	Docker, GitHub Actions, self-hosted Linux
Practices	Domain-Driven Design, Role-based access control

EDUCATION

University of Melbourne – Master of Software Engineering	Grade H2A	2023.07 – 2026.06
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